

Project assignment

Modelling and Simulation of Biological Systems

Assignment

This project aims to apply the principles of the course Modelling and Simulation to a *small*, self-contained example related to bioscience engineering. To this end, you can draw inspiration from one of our abstracts. You must work on your project in the Pluto notebook environment in the provided template (Dutch or English).

Your project must be of a maximum length of **eight to ten pages when printed**. Your project contains the following:

- ☐ an abstract with context, why it is relevant, a summary of what you have done and a conclusion
- ☐ a model based on differential equations or a probabilistic model (a stochastic program)
- ☐ a clear outline of the variables and parameters
- ☐ it has some aspect of either process optimization (so you use optimization to improve a parameter, input or decision), calibration (you tweak a parameter based on sampling or optimization) or gains some deeper insight in the system (e.g., how inputs influence the output).
- ☐ you use some form of uncertainty assessment, sensitivity analysis or some other stochastic component

These aspects can be extended or limited, as you choose. For example, you use your model for some process optimization via an optimizer but you can also tune a parameter by hand.

You must submit the project as a PDF, HTML, and Julia (.jl) file through UFORA **by 8 May**.

Each project should also end with a small attribution who did what parts according to the [CRediT](#) (Contribution Roles Taxonomy) classification. For example:

MS: Conceptualization, Methodology, Writing – Review & Editing; BP: Software, visualization, Formal Analysis; DG: Writing – Original Draft Preparation.

Rubric for grading

Category	0-1 Points (Unsatisfactory)	2 Points (Developing)	3 Points (Satisfactory)	4 Points (Good)	5 Points (Excellent)
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Clarity and Organization of Notebook	Notebook is very difficult to follow. Code and explanations are disorganized or missing.	Notebook has some organization, but it's challenging to understand the logic and purpose.	Notebook follows a generally clear structure with basic explanations of code and results.	Notebook is well-organized, with clear sections and explanations that guide the reader's understanding.	Notebook is exceptionally well-structured, with detailed comments and explanations that make it effortless to follow the project's logic.
Quality of Mathematical Model	Model is irrelevant to the chosen biological phenomenon or has major conceptual flaws.	Model shows some relevance to the problem but has significant simplifications or inaccuracies.	Model accurately captures the essential aspects of the biological phenomenon.	Model demonstrates good understanding of the system and includes relevant details and assumptions.	Model is sophisticated and incorporates nuanced or insightful elements that reflect a deep understanding of the biological system.
Quality of Analysis	Analysis is absent or uses incorrect techniques. Results are not presented or interpreted.	Analysis is attempted but flawed (errors, inappropriate methods). Results are presented with limited interpretation.	Analysis uses appropriate techniques and produces mostly correct results with basic interpretation.	Analysis employs suitable techniques leading to correct and meaningful results. Interpretation offers some insights.	Analysis utilizes a range of techniques providing a comprehensive understanding of the model behavior. Interpretation offers significant insights and implications.

In addition, 1/4 of the grade is determined by the oral defense.

Tips and advice

- Keep it as simple as possible. A project exploring a small model with two or three variables, outlining a well-known concept from your courses can lead to excellent projects or marks.

- You can use generative AI tools to help you at any point of the project, from brainstorming the initial idea to writing and proofreading text to helping with the code. You still have to explain everything and make sure the text looks "natural".
- Make sure you explain your reasoning well. Let others read it if it is clear how to do it.
- Make uninteresting or hard-to-understand pieces of code invisible if they don't help the reader.
- We expect figures to be tidy (titles, labels and axis). See the example projects for how to do this.
- Use clear variable names and use comments (#) to annotate parts that are not clear.
- The programming can and should be minimal.
- Teaching staff will be available for feedback after the lectures and labs or at designated time slots.
- Keep it as simple as possible!